

When the Curve Splits

Stress-testing a \$100M fixed income book against a non-parallel steepener and credit repricing

Dhanish Kandhari · 17 August 2026 · Market data as at 14 August 2026 close

Summary

On 29 July the FOMC under Kevin Warsh held rates and withheld forward guidance. Three officials dissented in favour of a hike. The market's answer was immediate and asymmetric: by 31 July the 30-year yield had reached **5.28%, a nineteen-year high**, while the 2-year barely moved.

Two weeks later the macro data had turned decisively softer. July payrolls printed **-23,000** against +83,000 expected, with 103,000 of downward revisions to prior months. September hike odds fell from roughly 55% to 42%. CPI cooled to 3.4%. The front end rallied hard.

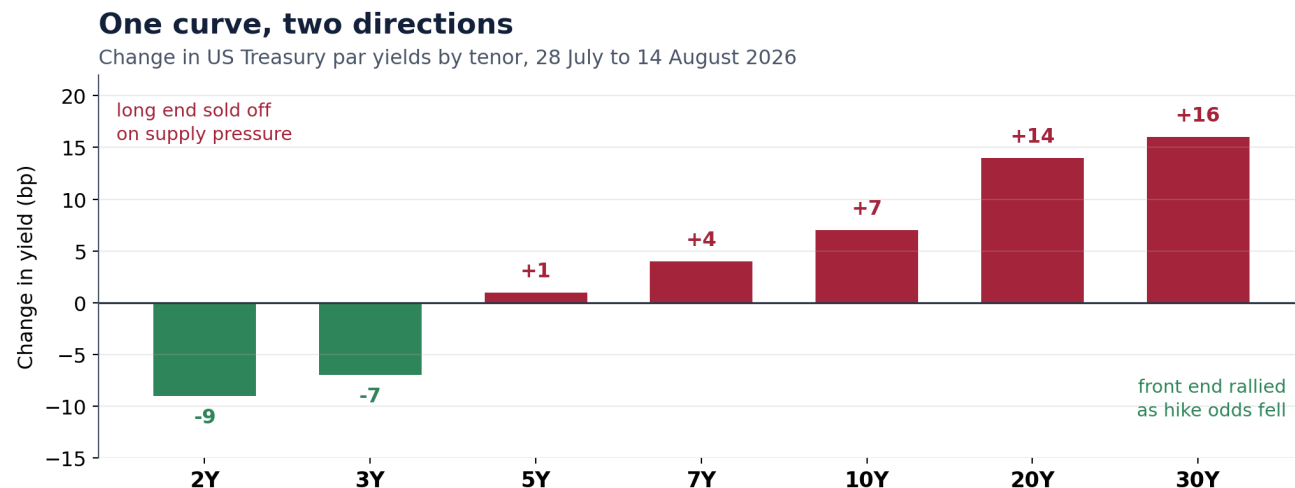
The long end did not follow. On 14 August the 30-year still stood at 5.25%.

That divergence is the substance of this note. A weakening labour market and cooling inflation should pull the entire curve lower. They pulled the front end only. The long end has stopped trading inflation expectations alone and started trading supply: heavy Treasury issuance, and concern that Japanese currency intervention involves selling US Treasuries. The steepening persisted across both halves of the period, but the driver rotated mid-move.

“When the short end and the long end stop telling the same story, standard risk assumptions break.”

This note quantifies what that environment does to a conventional \$100M multi-manager fixed income allocation. The tool is a Python risk engine that retrieves live prices and official effective durations from the fund provider, then applies non-parallel curve shocks alongside differentiated credit spread widening across three scenarios.

The realised move



Source: US Treasury daily par yield curve. The 5-year was the pivot of the move, effectively unchanged while both wings repriced in opposite directions.

Figure 1: The front end and the long end repriced in opposite directions over the same fortnight. 2s30s widened from +83 bp to +108 bp.

Portfolio and Scenario Design

The book is a conventional institutional allocation expressed through liquid ETF proxies. Each holding is mapped to the segment of the curve that drives it, so a non-parallel shift hits the three positions unevenly rather than uniformly.

Holding	Asset class	Weight	Capital	Eff. duration	Curve node
TLT	Long US Treasuries	40%	\$40.0M	14.92 yrs	Long end
LQD	US investment grade	45%	\$45.0M	7.74 yrs	Belly
HYG	US high yield	15%	\$15.0M	2.98 yrs	Short end
Portfolio		100%	\$100.0M	9.90 yrs	

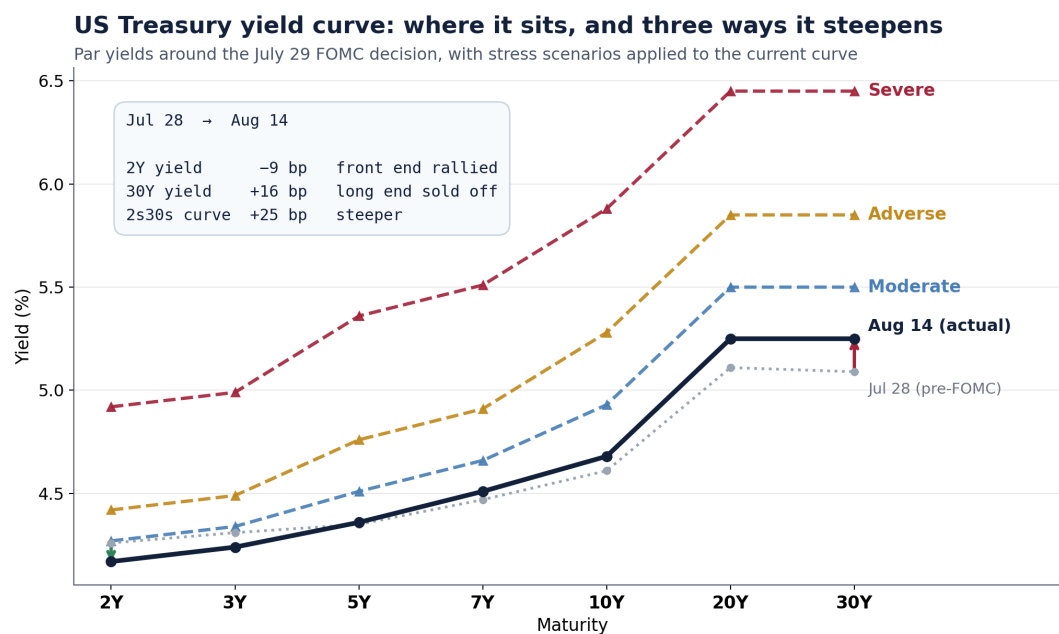
Effective durations retrieved from iShares, as at 13 August 2026. TLT carries no credit spread exposure.

Scenario calibration

Shock sizes are analyst assumptions, not retrieved data. They preserve the shape of the realised July-August move rather than imposing a parallel shift, and scale it. Spread shocks are differentiated by credit quality: high yield spreads historically move three to four times investment grade in risk-off episodes.

Scenario	Short	Belly	Long	IG spread	HY spread
Moderate	+10 bp	+15 bp	+25 bp	+25 bp	+75 bp
Adverse	+25 bp	+40 bp	+60 bp	+75 bp	+250 bp
Severe	+75 bp	+100 bp	+120 bp	+125 bp	+400 bp

LQD's option-adjusted spread stood at approximately 84 bp on 13 August against a long-run average near 160 bp. The Adverse IG shock therefore represents reversion to that average rather than a crisis. For scale, investment grade spreads reached roughly 400 bp in March 2020 and 600 bp in 2008, placing the Severe scenario between a 2022-style repricing and a genuine liquidity event.



Source: US Treasury daily par yield curve. Scenario shocks are analyst assumptions calibrated to the Jul 29 – Aug 14 move, applied at three curve nodes (short / belly / long), which is why the stressed lines move in parallel steps.

Figure 2: The current curve and the three stressed curves. Shocks are applied at three nodes, which is why the stressed curves move in parallel steps within each segment.

Methodology

Returns are computed as a first-order duration approximation applied as two independent legs. For a credit instrument the yield decomposes into a benchmark rate and an option-adjusted spread, each of which can move independently:

$$\text{return} = -(\text{effective duration} \times \text{rate shock}) - (\text{effective duration} \times \text{spread shock})$$

The spread leg is zero for Treasuries. Effective duration is used rather than modified duration because the underlying holdings contain embedded options - the majority of US high yield bonds are callable - and effective duration is computed by repricing under curve shifts while allowing cash flows to change. Consistently, spread shocks are expressed as movements in option-adjusted spread rather than nominal spread.

Results

Scenario	TLT	LQD	HYG	Portfolio	P&L	Closing
Moderate	-3.73%	-3.10%	-2.53%	-3.27%	-\$3.3M	\$96.7M
Adverse	-8.95%	-8.90%	-8.20%	-8.82%	-\$8.8M	\$91.2M
Severe	-17.90%	-17.42%	-14.16%	-17.12%	-\$17.1M	\$82.9M

Decomposition, Adverse scenario

Holding	Rate leg	Spread leg	Total	Dollar P&L	Weight after
TLT	-8.95%	0.00%	-8.95%	-\$3,580,800	39.9%
LQD	-3.10%	-5.80%	-8.90%	-\$4,005,450	45.0%
HYG	-0.74%	-7.45%	-8.20%	-\$1,229,250	15.1%

Weighted portfolio duration moves from 9.90 to 9.89 years. Losses are sufficiently uniform that the book does not passively de-risk.

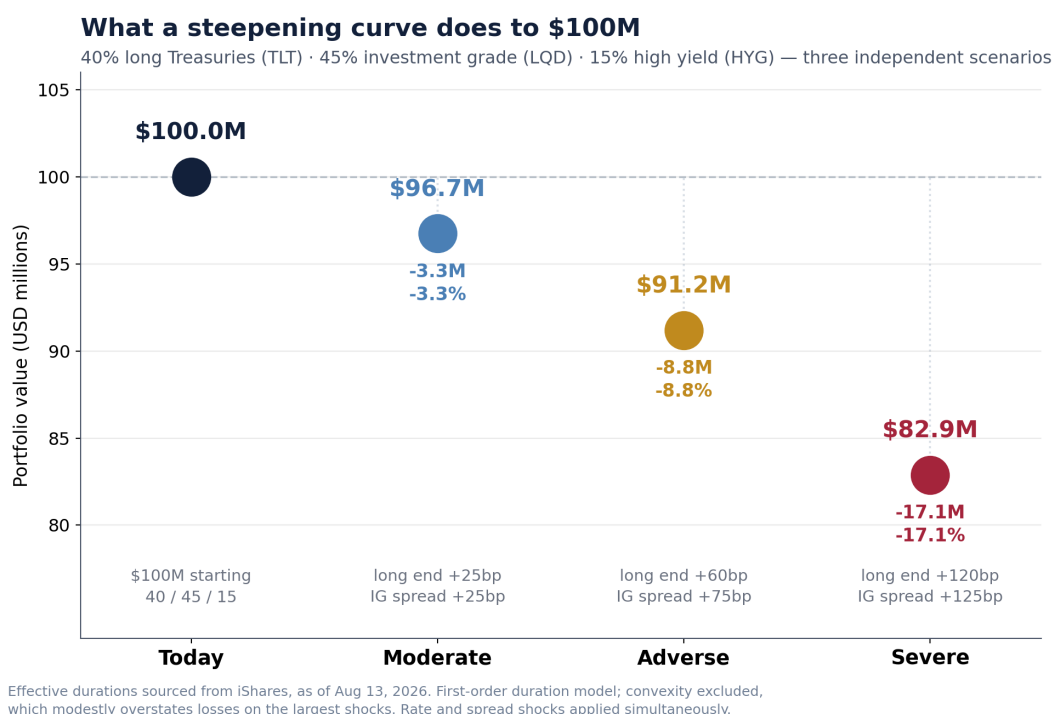


Figure 3: Closing portfolio value under each scenario. The three scenarios are independent states, not a sequence.

Findings

1. Asset-class diversification provided no protection

Under the Adverse scenario all three holdings landed within 75 bp of one another: -8.95%, -8.90% and -8.20%. Three entirely different credit profiles, one outcome. Diversification works when risk factors move independently. Here rates and spreads moved together, and every position in the book was exposed to at least one of them.

2. Identical losses, entirely different mechanisms

TLT's loss was pure duration. LQD's divided roughly evenly between the rate and spread legs. HYG's was almost entirely spread widening, its 2.98-year duration absorbing the rate shock almost completely. This matters for hedging: a duration overlay would have protected TLT and half of LQD, and done nothing at all for HYG. Same outcome, incompatible treatment.

3. Position size drove dollar risk, not volatility

LQD posted the largest dollar loss at \$4.0M against TLT's \$3.6M, despite the smaller percentage move, because it was the largest allocation absorbing both shocks. Percentage sensitivity ranks risk; position size determines it.

Taken together: the shock is a rates event that becomes a credit event, and a book constructed for diversification across credit quality offers little defence against it. The lever that matters is duration, and it must be pulled deliberately rather than left to the market.

Limitations

The model is deliberately first-order. Convexity is excluded, which overstates losses on the largest shocks; published convexity figures imply a true Severe loss nearer 15.3% than the 17.1% reported. That correction is a cushion for TLT and LQD but works in the opposite direction for HYG, whose convexity is negative. Effective duration proxies spread duration. Each holding maps to a single curve node. Shocks are instantaneous, with no coupon income offsetting the capital loss.

This is a scenario stress test, not a value-at-risk model and not a forecast. No probabilities are attached to any scenario.

Known extensions not implemented: convexity as a second-order term; separate spread duration inputs; a full key-rate duration vector rather than three nodes; carry and roll-down over a defined horizon; and historical scenario replay against the 2013 taper, March 2020 and full-year 2022 as a cross-check on the assumed shock sizes.

Data sourcing

Prices are retrieved per run via yfinance. Effective duration resolves through three layers: an audited manual override for the provider's daily figure, which is rendered client-side and unreachable by a requests-based scraper; automated parsing of the official fact sheet PDF; and product-page HTML parsing anchored on the visible label. Every resolved duration is range-checked before entering the calculation, and an out-of-range value halts the run rather than corrupting the result. No investment characteristics are hardcoded.

Full source code, methodology and the complete assumptions register are available on [GitHub](#). Prepared for discussion; not investment advice.